DESIGN COMPUTED · FREE GUIDE

Never four corners together: the tatami rule and the module behind a Japanese room

The free guide to the reel on tatami mats: how rooms are measured in mats, why plus-shaped junctions are avoided, how regional mat sizes differ, and a paper-mat tiling exercise.



A floor made of one module

A tatami mat is about 0.9 by 1.8 m, twice as long as wide. The surface is a weft-faced weave of igusa rush on a hemp or cotton warp, and the core is traditionally rice straw, though modern ones may use wood chip board or foam.

Rooms are sized in mats (jo). A square of two mats is a tsubo, about 3.3 square metres. A half mat is a hanjo, and a three-quarter mat is used in tea rooms.

Mat sizes differ by region

Kyoma (Kyoto)	1.91 by 0.955 m, about 5.5 cm thick.
Chukyoma (Nagoya)	1.82 by 0.91 m.
Edoma / Kantoma (Tokyo)	1.76 by 0.88 m, about 6.0 cm thick.

The rule: no plus-shaped junctions

Auspicious layouts stagger the mats so every junction is a T. They are the standard today. Layouts in a plain grid, where four corners meet at a plus (+), were used for funerals and other events and are avoided in ordinary rooms.

Auspicious layouts often need half mats. Checking whether a large room can be covered with an auspicious layout using only full mats is a computationally hard problem (NP-complete).

Built examples

A 6-mat room (Nagoya size)	12 by 9 shaku, about 3.64 by 2.73 m.
A 4.5-mat room	9 by 9 shaku, about 2.73 by 2.73 m, the classic small room.
An 8-mat room	12 by 12 shaku, about 3.64 by 3.64 m.

Try it yourself: tile a 6-mat room

1. Cut six rectangles of paper, each 2 by 1 units (for example 10 by 5 cm).
2. Draw a room outline 4 units wide and 3 units tall.
3. Lay the six mats in a plain grid and mark every point where four corners meet. This is the funeral layout.
4. Rearrange the same mats so no point has four corners (turn two mats upright). Mark every T-junction.
5. Now try a 4.5-mat room, which needs a half mat. How many different lucky layouts can you find?

The rule to remember

A module plus a rule gives you a whole building system, and the rule can be checked by a computer.

Default answer	Draw each room from scratch.
Form-finder answer	Choose one module and one rule, then let the layouts follow.
Check	Ask: what is the mat in my project, and what is its plus-junction rule?

Go further

Tatami (Wikipedia)
https://en.wikipedia.org/wiki/Tatami

Japanese-style room, washitsu (Wikipedia)
https://en.wikipedia.org/wiki/Washitsu

Tatami tiling problem (Wikipedia)
https://en.wikipedia.org/wiki/Tatami

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